

Part 1: unsigned number (integer) representation

Exercise 1

- Give the number of values and the representation interval of unsigned integers on 4, 8, 16 and 32 bits.
- How many minimum bits are required to represent the following numbers as unsigned integers:
 - A. 14_{10}
 - B. 38_{10}
 - C. 15_{10}

Part 2: signed number (integer) representation

Exercise 2

- Give on 8 bits the “sign-magnitude”, “one’s complement” and “two’s complement” representations of the numbers 45, 73, -102, and -118.
- Decode the following 8-bit two's Complement-encoded signed integers :
 - A. $1100\ 1001_2$
 - B. $0110\ 1101_2$
 - C. $1000\ 0000_2$

Exercise 3

Give the decimal value of the binary numbers 1111 1111, 1001 0001 each considered in an unsigned 8-bit integer encoding, or signed by sign-magnitude, or signed in one's complement, and finally, signed in two's complement .

Exercise 4

1. Encode $(61)_{10}$ and $-(61)_{10}$ in one byte using sign and absolute value representation.
2. Add $(61)_{10}$ and $-(61)_{10}$ in sign and absolute value (sign-magnitude).
3. Encode $(61)_{10}$ and $-(61)_{10}$ on one byte using two's Complement representation (C2).
4. Add $(61)_{10}$ and $-(61)_{10}$ in two's Complement. What is your conclusion?

Exercise 5

For 8-bit two's Complement coding:

1. What is the largest positive number (in decimal and binary)
2. What is the largest negative number (in decimal and binary)
3. What is the smallest negative number (in binary)

Exercise 6

1. perform the following arithmetic operations :
 - A. $52 + 27$ en binaire.
 - B. $7,25 + 2,5$ en binaire
 - C. $(11100)_2 - (11011)_2$
 - D. $(1100)_2 - (0110)_2$
 - E. $(11110)_2 * (110)_2$
 - F. $(1011110)_2 / (101)_2$
2. Perform the following operations in tow's complement (on 8 bits):
 - A. $1_{10} - 2_{10}$;
 - B. $51_{10} + 127_{10}$;
 - C. $3_{10} - 127_{10}$;
 - D. $-122_{10} + 43_{10}$

Specify the carry and overflow for each operation.

Part 2: representation of characters

Exercise 7

A book has an average of 800 pages. If each page is made up of 90 columns and 50 rows, how many medium sized books can be stored on an 8 GB size USB memory?

Exercise 8

- What does the following binary sequence represent if it is encoded in ASCII:
1010011101010010011111010000
- Which bit must be complemented to change an uppercase ASCII letter into a lowercase ASCII letter and vice versa?